

Nationwide House Energy Rating Scheme®

NatHERS® Certificate No. 09MUU5PPKV

Generated on 13 Jan 2026 using FirstRate5: 5.5.5a (3.22)

Property

Address 16 Zillie Close ,
Westdale, NSW, 2830
Lot/DP 1808/DP1318131
NCC Class* Class 1a
Floor/all Floors
Type New Home

Plans

Main plan 11439
Prepared by HIBBARDS PTY LTD

Construction and environment

Assessed floor area [m²]*	Exposure type
Conditioned* 143.4	suburban
Unconditioned* 47.4	NatHERS climate zone
Total 190.8	48 Dubbo Airport
Garage 34.4	



Accredited assessor

Name SANTARITA KOLLOSCHÉ
Business name Stellar Thermal Assessments
Email admin@stellarthermal.com
Phone +61481099915
Accreditation No. HERA10183
Assessor Accrediting Organisation
HERA
Declaration of interest Yes, managed

NCC Requirements

NCC provisions Volume 2
State/Territory variation Yes

National Construction Code (NCC) requirements

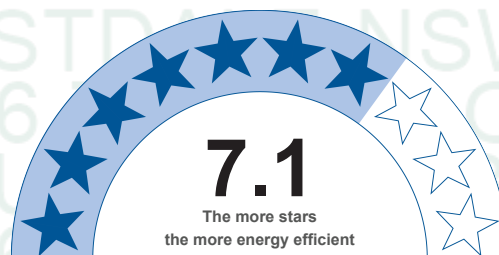
The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J3D3 and J3D15 of NCC Volume One.

NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.

The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au.

Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.

Thermal performance star rating



**NATIONWIDE
HOUSE**
ENERGY RATING SCHEME®

77.9 MJ/m²

Predicted annual energy load for heating and cooling based on standard occupancy assumptions.

For more information on your dwelling's rating see:
www.nathers.gov.au

Thermal performance [MJ/m²]

Limits taken from ABCB Standard 2022

	Heating	Cooling
Modelled	60.8	17.1
Load limits	N/A	N/A
Features determining load limits		
Floor type (lowest conditioned area)		N/A
NCC climate zone 1 or 2		N/A
Outdoor living area		N/A
Outdoor living area ceiling fan		N/A

Whole of Home performance rating

No Whole of Home performance rating generated for this certificate

Verification

To verify this certificate, scan the QR code or visit <https://www.fr5.com.au/QRCodeLandIng?PublicId=09MUU5PPKV>. When using either link, ensure you are visiting www.fr5.com.au.





About the ratings

Thermal performance rating

NatHERS thermal software models the expected heating and cooling energy loads using information about the design, construction, climate and common patterns of household use. The thermal performance rating (shown as a star rating on this Certificate) does not take into account appliances, apart from the airflow impacts from ceiling fans.

Whole of Home performance rating

NatHERS Whole of Home software uses the heating and cooling energy loads combined with the energy performance of the home's appliances (heating, cooling, hot water, lighting, pool/spa pump and onsite renewable energy generation and storage) and models the expected energy value* of the whole home. The Whole of Home performance rating is shown as a score out of 100 on this Certificate.

Heating & Cooling Load Limits

Additional information

In some locations under the NCC NatHERS pathway, separate heating and cooling load limits may apply. Minimum required star ratings in northern parts of Australia may also be affected by the presence or absence of an outdoor living area and/or an outdoor living area ceiling fan. Refer to the ABCB NatHERS heating and cooling load limits Standard 2022 for details or contact the relevant local building regulating authority, noting that State and Territory variations may also apply.

Setting options:

Floor type:

- CSOG – Concrete Slab on Ground
- SF – Suspended Floor (or a mixture of CSOG and SF)
- NA – Not Applicable

NCC climate Zone 1 or 2:

- Yes
- No
- NA – not applicable

Outdoor living area:

- Yes
- No
- NA – not applicable

Outdoor living area ceiling fan:

- Yes
- No
- NA – not applicable

Predicted Whole of Home annual impact by appliance

Shows the contribution each appliance has on the home's annual energy use, greenhouse gas emissions and cost without solar

Energy use:

No Whole of Home performance assessment conducted for this certificate.

Greenhouse gas emissions:

No Whole of Home performance assessment conducted for this certificate.

Cost:

No Whole of Home performance assessment conducted for this certificate.

Graph key:



Predicted onsite renewable energy impact

No Whole of Home performance assessment conducted for this certificate.



Certificate check

The checklist covers important items impacting the dwelling's ratings.
It is recommended that the accuracy of the whole certificate is checked.

Note: The boxes indicate when and who should check each item.
It is not mandatory to complete this checklist.

	Approval stage		Construction stage		Occupancy/other
	Assessor checked	Consent authority/surveyor checked	Builder checked	Consent authority/surveyor checked	
Genuine certificate check					
Does this Certificate match the one available at the web address or QR code verification link on the front page?		☐	☐	☐	☐
Does the NatHERS certificate number on the NatHERS-stamped plans match the number on this Certificate?		☐	☐	☐	☐
Thermal performance check					
Windows and glazed doors					
Does the window size, opening type and location shown on the NatHERS-stamped plans or as installed match what is shown in 'Window and glazed door schedule' and 'Roof window schedule' tables on this Certificate?	☐	☐	☐	☐	☐
Does the installed windows meet the substitution tolerances (AFRC* based SHGC* and U-values*) as shown in the 'Window and glazed door type and performance' and 'Roof window type and performance' tables on this Certificate?			☐	☐	☐
External walls					
Does the external wall bulk insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the External wall type table on this Certificate?	☐	☐	☐	☐	☐
Does the external wall shade (colour) match what is shown in the 'External wall type' table on this Certificate?	☐	☐	☐	☐	☐
Floor					
Does the floor insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Floor type' table on this certificate?	☐	☐	☐	☐	☐
Ceiling penetrations*					
Does the 'quantity' and 'type' of ceiling penetrations* (e.g. downlights, exhaust fans, etc) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling penetrations' table on this Certificate?	☐	☐	☐	☐	☐
Ceiling					
Does the ceiling insulation (R-value) shown on the NatHERS-stamped plans or as installed match what is shown in the 'Ceiling type' table on this Certificate?	☐	☐	☐	☐	☐
Roof					
Does the external roof shade (colour) on the NatHERS stamped plans or as installed match what is shown in the 'Roof type' table on this Certificate?	☐	☐	☐	☐	☐
Apartment entrance doors (NCC Class 2 assessments only)					
Does the 'External Door Schedule' show apartment entrance doors? Please note that an "external door" between the modelled dwelling and a shared space, such as an enclosed corridor or foyer, should not be included in the assessment (because it overstates the possible ventilation) and would invalidate the Certificate.	☐	☐		☐	☐
Exposure*					
Has the appropriate exposure type (terrain) (shown on page 1) been applied? For example, it is unlikely that a ground-floor apartment is "exposed" or a top floor high-rise apartment is "protected".	☐	☐		☐	☐
Heating and cooling load limits*					
Do the load limits settings (shown on page 1) match the values in the ABCB Standard 2022: NatHERS heating and cooling load limits for the appropriate climate zone?	☐	☐	☐	☐	☐



Certificate check

Continued

	Approval stage		Construction stage		
	Assessor checked	Consent authority/ surveyor checked	Builder checked	Consent authority/ surveyor checked	Occupancy/other
Additional NCC requirements for thermal performance (not included in the NatHERS assessment)					
Thermal bridging					
Does the dwelling meet the NCC requirement for thermal bridging?		☐	☐	☐	☐
Insulation installation method					
Has the insulation been installed according to the NCC requirements?			☐	☐	☐
Building sealing					
Does the dwelling meet the NCC requirements for Building Sealing?		☐	☐	☐	☐
Whole of Home performance check (not applicable if a Whole of Home performance assessment is not conducted)					
Appliances					
Does the cooling appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the Appliance schedule on this Certificate?	☐	☐	☐	☐	☐
Does the heating appliance/s type, location and efficiency/performance shown on the NatHERS-stamped plans or installed, match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the hot water system type and efficiency/performance shown on the NatHERS-stamped plans or as installed match the location and minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the pool pump efficiency/performance shown on the NatHERS-stamped plans or as installed match the minimum efficiency/performance requirements shown in the 'Appliance schedule' on this Certificate?	☐	☐	☐	☐	☐
Does the onsite renewable energy system type, orientation and system size or generation capacity shown on the NatHERS stamped plans or installed match the 'Onsite Renewable Energy schedule' on this Certificate?	☐	☐	☐	☐	☐
Additional NCC Requirements for Services (not included in the NatHERS assessment)					
Does the lighting meet the artificial lighting requirements specified in the NCC?		☐	☐	☐	☐
Does the hot water system meet the additional requirements specified in the NCC?		☐	☐	☐	☐
Provisional values* check					
Have provisional values* been used in the assessment and, if so, are they noted in 'Additional notes' table below?	☐	☐			

Other NCC requirements

Note: This Certificate only covers the energy efficiency requirements in the NCC. Additional requirements that must also be satisfied include, but are not limited to: condensation, structural and fire safety requirements and any state or territory variations to the NCC energy efficiency requirements.

Additional notes

R2.0 HIGH DENSITY INSULATION BATTS TO INTERNAL WALLS JOINING TO GARAGE ONLY

R2.0 INSULATION BATTS W/ NR WRAP TO ALL EXTERNAL WALLS INCLUDING GARAGE

R6.0 INSULATION BATTS TO ENTIRE CEILING INCLUDING GARAGE / EDGE BATTS INCLUDED

R1.3 ANTICON BLANKET TO ENTIRE CEILING INCLUDING GARAGE

GLAZING: SINGLE CLEAR / COMFORT PLUS (W5,SD4) / DOUBLE GLAZED (W6,W9)

CEILING FAN: N/A



Room *schedule*

Room	Zone Type	Area [m ²]
GARAGE	garage	34.4
LAUNDRY	unconditioned	4.4
KIT/FAM/DIN	kitchen	40.6
BED 4	bedroom	12.3
BED 2	bedroom	11.6
BED 3	bedroom	11
WIR	nightTime	4.4
MEDIA	dayTime	13.5
BATHROOM	unconditioned	7.1
WC	unconditioned	1.5
ENSUITE	nightTime	5.6
BED 1	bedroom	14
ENTRY / HALL	dayTime	11.8
ACTIVITY	living	19.3

Window and glazed door *type and performance*

Default* windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
No Data Available					

Custom* windows

Window ID	Window description	Maximum U-value*	SHGC*	Substitution tolerance ranges	
				SHGC lower limit	SHGC upper limit
WID-012-01 A	Aluminium Awning Window SG 5mmClr	6.32	0.63	0.6	0.66
WID-005-01 A	Al Residential Internal Sliding Door SG 4mm Clear	6.25	0.72	0.68	0.76
WID-006-01 A	Al Residential Sliding Window SG 3mm Clear	6.42	0.76	0.72	0.8
WID-005-04 A	Al Residential Internal Sliding Door SG 638mm Comfort Plus	4.42	0.44	0.42	0.46
WID-006-04 A	Al Residential Sliding Window SG 638mm Comfort Plus	4.53	0.46	0.44	0.48
WID-006-09 A	Al Residential Sliding Window DG 3mm Clear / 6mm Air Gap / 4mm Energy Advanta	3.99	0.56	0.53	0.59



WID-028-09 A	BSW Ascend Sliding Window SG 5mmClr	6.55	0.62	0.59	0.65
WID-003-01 A	AI Residential Double Hung Window SG 3Clr	6.24	0.74	0.7	0.78

Window and glazed door *schedule*

Location	Window ID	Window no.	Height [mm]	Width [mm]	Window type	Opening %	Orientation	Window shading device*
GARAGE	WID-012-01 A	W1	300	1800	fixed	0.0	N	No
LAUNDRY	WID-005-01 A	D3	2100	1500	sliding	45.0	N	No
KIT/FAM/DIN	WID-006-01 A	W2	1200	1500	sliding	45.0	N	No
KIT/FAM/DIN	WID-005-04 A	D4	2100	2400	sliding	45.0	W	No
BED 4	WID-006-01 A	W4	1500	1500	sliding	45.0	W	No
BED 2	WID-006-01 A	W3	1800	1200	sliding	45.0	N	No
BED 3	WID-006-04 A	W5	1500	1500	sliding	45.0	W	No
MEDIA	WID-006-09 A	W9	1200	2100	sliding	45.0	S	No
BATHROOM	WID-028-09 A	W8	900	1200	sliding	45.0	S	No
WC	WID-028-09 A	W7	900	600	sliding	45.0	S	No
ENSUITE	WID-028-09 A	W10	900	1200	sliding	45.0	S	No
BED 1	WID-003-01 A	W11	1800	900	double_hung	45.0	E	No
BED 1	WID-003-01 A	W12	1800	900	double_hung	45.0	E	No
ACTIVITY	WID-006-09 A	W6	1200	2100	sliding	45.0	S	No

Roof window* *type and performance value*

Default* roof windows

				Substitution tolerance ranges	
Window ID	Window description	Maximum U-value*	SHGC*	SHGC lower limit	SHGC upper limit
No Data Available					

Custom* roof windows

				Substitution tolerance ranges	
Window ID	Window description	Maximum U-value*	SHGC*	SHGC lower limit	SHGC upper limit
No Data Available					

Roof window* *schedule*

Location	Window ID	Window no.	Opening %	Area [m²]	Width [mm]	Orientation	Outdoor shade	Indoor shade
No Data Available								

Skylight* *type and performance*

Skylight ID	Skylight description	Skylight shaft reflectance
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No Data Available

Skylight* *schedule*

Location	Skylight ID	Skylight No.	Skylight shaft length [mm]	Area [m²]	Orient-ation	Outdoor shade	Diffuser
No Data Available							

External door *schedule*

Location	Height [mm]	Width [mm]	Opening %	Orientation
GARAGE	2100	4800	0.0	E
ENTRY / HALL	2040	1200	100.0	E

External wall *type*

Wall ID	Wall type	Solar absorptance	Wall shade [colour]	Bulk insulation [R-value]	Reflective wall wrap*
1	STA - HARDIES AXON 90mm STUD+BULK	0.3	Light	Glass fibre batt: R2.0 (R2.0)	No

External wall *schedule*

Location	Wall ID	Height [mm]	Width [mm]	Orientation	Horizontal shading feature* maximum projection [mm]	Vertical shading feature* (yes/no)
GARAGE	1	2526	5989	N	553	Yes
GARAGE	1	2526	552	E	570	Yes
GARAGE	1	2526	5191	E	600	Yes
LAUNDRY	1	2440	2877	N	0	Yes
KIT/FAM/DIN	1	2440	4509	N	554	Yes
KIT/FAM/DIN	1	2440	3522	W	5688	Yes
BED 4	1	2440	3246	N	4150	Yes
BED 4	1	2440	3985	W	551	Yes
BED 2	1	2440	1484	N	0	Yes
BED 2	1	2441	334	N	4177	Yes
BED 3	1	2440	3587	W	550	Yes
BED 3	1	2440	3247	S	556	Yes
MEDIA	1	2440	3599	S	550	Yes
BATHROOM	1	2440	1789	S	564	Yes
WC	1	2440	893	S	584	Yes
ENSUITE	1	2440	598	W	0	Yes
ENSUITE	1	2440	2041	S	0	Yes
ENSUITE	1	2440	609	E	0	No
BED 1	1	2442	254	E	400	No



BED 1	1	2441	3498	E	400	Yes
BED 1	1	2440	1303	N	1650	Yes
BED 1	1	2440	3744	S	0	Yes
ENTRY / HALL	1	2440	1594	E	1721	Yes
ACTIVITY	1	2440	4143	S	557	Yes

Internal wall type

Wall ID	Wall type	Area [m²]	Bulk insulation
1	FR5 - Internal Plasterboard Stud Wall	28.7	Glass fibre batt: R2.0 (R2.0)
2	FR5 - Internal Plasterboard Stud Wall	150.6	

Floor type

Location	Construction	Area [m²]	Sub-floor ventilation	Added insulation [R-value]	Covering
GARAGE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	2.5	Enclosed	R0.0	none
GARAGE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	29.6	Enclosed	R0.0	none
GARAGE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	2.1	Enclosed	R0.0	none
GARAGE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.2	Enclosed	R0.0	none
LAUNDRY	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	3.3	Enclosed	R0.0	Tiles
LAUNDRY	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.1	Enclosed	R0.0	Tiles
KIT/FAM/DIN	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	37.5	Enclosed	R0.0	Tiles
KIT/FAM/DIN	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.4	Enclosed	R0.0	Tiles
KIT/FAM/DIN	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.2	Enclosed	R0.0	Tiles
KIT/FAM/DIN	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.6	Enclosed	R0.0	Tiles
BED 4	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	9.7	Enclosed	R0.0	Carpet
BED 4	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.5	Enclosed	R0.0	Carpet
BED 4	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.1	Enclosed	R0.0	Carpet
BED 4	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1	Enclosed	R0.0	Carpet
BED 2	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.8	Enclosed	R0.0	Carpet



BED 2	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	10.8	Enclosed	R0.0	Carpet
BED 3	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1	Enclosed	R0.0	Carpet
BED 3	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.2	Enclosed	R0.0	Carpet
BED 3	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.4	Enclosed	R0.0	Carpet
BED 3	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	8.4	Enclosed	R0.0	Carpet
WIR	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	4.4	Enclosed	R0.0	Carpet
MEDIA	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.3	Enclosed	R0.0	Carpet
MEDIA	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	12.2	Enclosed	R0.0	Carpet
BATHROOM	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.7	Enclosed	R0.0	Tiles
BATHROOM	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	6.4	Enclosed	R0.0	Tiles
WC	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.3	Enclosed	R0.0	Tiles
WC	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.2	Enclosed	R0.0	Tiles
ENSUITE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.7	Enclosed	R0.0	Tiles
ENSUITE	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	4.8	Enclosed	R0.0	Tiles
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.2	Enclosed	R0.0	Carpet
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.6	Enclosed	R0.0	Carpet
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	10.4	Enclosed	R0.0	Carpet
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.2	Enclosed	R0.0	Carpet
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.4	Enclosed	R0.0	Carpet
BED 1	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.2	Enclosed	R0.0	Carpet
ENTRY / HALL	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	11.1	Enclosed	R0.0	Tiles
ENTRY / HALL	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	0.7	Enclosed	R0.0	Tiles
ACTIVITY	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	1.5	Enclosed	R0.0	Carpet



ACTIVITY	FR5 - 300mm waffle pod, 85mm concrete (R0.63)	17.8	Enclosed	R0.0	Carpet
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Ceiling type

Location	Construction material/type	Bulk insulation R-value [may include edge batt values]	Reflective wrap*
GARAGE	Plasterboard	R3.0	Yes
GARAGE	Plasterboard	R6.0	Yes
GARAGE	Plasterboard	R3.0	Yes
LAUNDRY	Plasterboard	R6.0	Yes
LAUNDRY	Plasterboard	R3.0	Yes
KIT/FAM/DIN	Plasterboard	R6.0	Yes
KIT/FAM/DIN	Plasterboard	R3.0	Yes
KIT/FAM/DIN	Plasterboard	R3.0	Yes
BED 4	Plasterboard	R6.0	Yes
BED 4	Plasterboard	R3.0	Yes
BED 4	Plasterboard	R3.0	Yes
BED 2	Plasterboard	R3.0	Yes
BED 2	Plasterboard	R6.0	Yes
BED 3	Plasterboard	R3.0	Yes
BED 3	Plasterboard	R3.0	Yes
BED 3	Plasterboard	R6.0	Yes
WIR	Plasterboard	R6.0	Yes
MEDIA	Plasterboard	R3.0	Yes
MEDIA	Plasterboard	R6.0	Yes
BATHROOM	Plasterboard	R3.0	Yes
BATHROOM	Plasterboard	R6.0	Yes
WC	Plasterboard	R6.0	Yes
ENSUITE	Plasterboard	R3.0	Yes
ENSUITE	Plasterboard	R6.0	Yes
BED 1	Plasterboard	R3.0	Yes
BED 1	Plasterboard	R3.0	Yes
BED 1	Plasterboard	R6.0	Yes
BED 1	Plasterboard	R3.0	Yes
ENTRY / HALL	Plasterboard	R6.0	Yes
ENTRY / HALL	Plasterboard	R3.0	Yes
ACTIVITY	Plasterboard	R3.0	Yes
ACTIVITY	Plasterboard	R6.0	Yes

Ceiling penetrations*



Location	Quantity	Type	Height [mm]	Width [mm]	Sealed/unsealed
LAUNDRY	1	Downlights	80	80	Sealed
KIT/FAM/DIN	6	Downlights	80	80	Sealed
KIT/FAM/DIN	1	Exhaust Fans	200	200	Sealed
BED 4	1	Downlights	80	80	Sealed
BED 2	1	Downlights	80	80	Sealed
BED 3	1	Downlights	80	80	Sealed
WIR	1	Downlights	80	80	Sealed
MEDIA	2	Downlights	80	80	Sealed
BATHROOM	1	Downlights	80	80	Sealed
BATHROOM	1	Exhaust Fans	200	200	Sealed
WC	1	Downlights	80	80	Sealed
ENSUITE	1	Downlights	80	80	Sealed
ENSUITE	1	Exhaust Fans	200	200	Sealed
BED 1	1	Downlights	80	80	Sealed
ENTRY / HALL	1	Downlights	80	80	Sealed
ACTIVITY	2	Downlights	80	80	Sealed

Ceiling fans

Location	Quantity	Diameter [mm]
No Data Available		

Roof type

Construction	Added insulation [R-value]	Solar absorptance	Roof shade [colour]
Cont:Attic-Continuous	1.3	0.44	Medium

Thermal bridging *schedule for steel frame elements*

Building element	Steel section dimensions [height x width, mm]	Frame spacing [mm]	Steel thickness [BMT,mm]	Thermal break [R-value]
No Data Available				

Appliance *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

Note: A flat assumption of 5W/m2 is used for lighting, therefore lighting is not included in the appliance schedule.

Cooling system

Appliance/ system type	Location	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.				



Heating system

Appliance/ system type	Location	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.				

Hot water system

Appliance/ system type	Fuel type	Minimum efficiency/ performance	Hot Water CER Zone	Zone 3 STC	Assessed daily load
No Whole of Home performance assessment conducted for this certificate.					

Pool/spa equipment

Appliance/ system type	Fuel type	Minimum efficiency/ performance	Recommended capacity
No Whole of Home performance assessment conducted for this certificate.			

Onsite renewable energy *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

System type	Orientation	System size or generation capacity
No Whole of Home performance assessment conducted for this certificate.		

Battery *schedule*

(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

System type	Size [battery storage capacity]
No Whole of Home performance assessment conducted for this certificate.	

Explanatory Notes

About this report

NatHERS ratings are a reliable guide for comparing different dwelling designs and to demonstrate that designs meet the energy efficiency requirements in the National Construction Code.

NatHERS ratings use computer modelling to evaluate a home's energy efficiency and performance. They use localised climate data and standard assumptions on how people use their home to predict the heating and cooling energy loads and energy value* of the whole home. The thermal performance star rating uses the home's building specifications, layout, orientation and fabric (i.e. walls, windows, floors, roofs and ceilings) to predict the heating and cooling energy loads. The Whole of Home performance rating uses information about the home's appliances and onsite energy generation and storage to estimate the homes energy value*.

The actual energy loads, cost and greenhouse gas emissions of a home may vary from that predicted. This is because the assumptions will not always match the actual occupant usage patterns. For example, the number of occupants and how people use their appliances will vary. Energy efficient homes use less energy, are warmer on cool days, cooler on hot days and cost less to run.

Accredited assessors

For quality assured NatHERS Certificates, always use an accredited or licenced assessor registered with an Assessor Accrediting Organisation (AAO). AAOs have strict quality assurance processes, and professional development requirements ensuring consistently high standards for assessments.

Non-accredited assessors (Raters) have no ongoing training requirements and are not quality assured.

Any queries about this report should be directed to the assessor. If the assessor is unable to address questions or concerns, contact the AAO specified on the front of this certificate.

Disclaimer

The NatHERS Certificate format is developed by the NatHERS Administrator. However, the content in the certificate is entered by the assessor. It is the assessor's responsibility to use NatHERS accredited software correctly and follow the NatHERS Technical Note to produce a NatHERS Certificate.

The predicted annual energy load, cost and greenhouse gas emissions in this NatHERS Certificate are an estimate based on an assessment of the dwelling's design by the assessor. It is not a prediction of actual energy use, cost or emissions. The information and ratings may be used to compare how other dwellings are likely to perform when used in a similar way.

Information presented in this report relies on a range of standard assumptions (both embedded in NatHERS accredited software and made by the assessor who prepared this report), including assumptions about occupancy, behaviour, appliance performance, indoor air temperature and local climate.

Not all assumptions made by the assessor using the NatHERS accredited software tool are presented in this report and further details or data files may be obtained from the assessor.

Glossary

Annual energy load	the predicted amount of energy required for heating and cooling, based on standard occupancy assumptions.
AFRC	Australian Fenestration Rating Council
Assessed floor area	the floor area modelled in the software for the purpose of the NatHERS assessment. Note, this may not be consistent with the floor area in the design documents.
Ceiling penetrations	features that require a penetration to the ceiling, including downlights, vents, exhaust fans, range hoods, chimneys and flues. Excludes fixtures attached to the ceiling with small holes through the ceiling for wiring, e.g. ceiling fans; pendant lights, and heating and cooling ducts.
Conditioned	a zone within a dwelling that is expected to require heating and cooling based on standard occupancy assumptions. In some circumstances it will include garages.
COP	Coefficient of performance
Custom windows	windows listed in NatHERS software that are available on the market in Australia and have a WERS (Window Energy Rating Scheme) rating.
Default windows	windows that are representative of a specific type of window product and whose properties have been derived by statistical methods.
EER	Energy Efficiency Ratio, measure of how much cooling can be achieved by an air conditioner for a single kWh of electricity input
Energy use	This is your homes rating without solar or batteries.
Energy value	The net cost to society including, but not limited to, costs to the building user, the environment and energy networks (as defined in the ABCB Housing Provisions Standard).
Entrance door	these signify ventilation benefits in the modelling software and must not be modelled as a door when opening to a minimally ventilated corridor in a Class 2 building.
Exposure category – exposed	terrain with no obstructions e.g. flat grazing land, ocean-frontage, desert, exposed high-rise unit (usually above 10 floors).
Exposure category – open	terrain with few obstructions at a similar height e.g. grasslands with few well scattered obstructions below 10m, farmland with scattered sheds, lightly vegetated bush blocks, elevated units (e.g. above 3 floors).
Exposure category – suburban	terrain with numerous, closely spaced obstructions below 10m e.g. suburban housing, heavily vegetated bushland areas.
Exposure category – protected	terrain with numerous, closely spaced obstructions over 10 m e.g. city and industrial areas.
Horizontal shading feature	provides shading to the building in the horizontal plane, e.g. eaves, verandahs, pergolas, carports, or overhangs or balconies from upper levels.
National Construction Code (NCC) Class	the NCC groups buildings by their function and use, and assigns a classification code. NatHERS software models NCC Class 1, 2 or 4 buildings and attached Class 10a buildings. Definitions can be found at www.abcb.gov.au .
Net zero home	a home that achieves a net zero energy value*.
Opening percentage	the openability percentage or operable (moveable) area of doors or windows that is used in ventilation calculations.
Provisional value	an assumed value that does not represent an actual value. For example, if the wall colour is unspecified in the documentation, a provisional value of 'medium' must be modelled. Acceptable provisional values are outlined in the NatHERS Technical Note and can be found at www.nathers.gov.au
Recommended capacity	this is the capacity or size of equipment that is recommended by NatHERS to achieve the desired comfort conditions in the zone or zones serviced. This is a recommendation and the final selection sizing should be confirmed by a suitably qualified person.
Reflective wrap (also known as foil)	can be applied to walls, roofs and ceilings. When combined with an appropriate air gap and emissivity value, it provides insulative properties.
Roof window	for NatHERS this is typically an operable window (i.e. can be opened), will have a plaster or similar light well if there is an attic space, and generally does not have a diffuser.
Shading features	includes neighbouring buildings, fences, and wing walls, but excludes eaves.
Solar heat gain coefficient (SHGC)	the fraction of incident solar radiation admitted through a window, both directly transmitted as well as absorbed and subsequently released inward. SHGC is expressed as a number between 0 and 1. The lower a window's SHGC, the less solar heat it transmits.
Skylight (also known as roof lights)	for NatHERS this is typically a moulded unit with flexible reflective tubing (light well) and a diffuser at ceiling level.

*Refer to glossary.



STCs	Small-scale Technology Certificates, certificates created by the REC registry for renewable energy technologies that may be bought and sold as part of the Small-scale Renewable Energy Scheme operated by the Clean Energy Regulatory
Thermal breaks	are materials with an R-value greater than or equal to 0.2 that must separate the metal frame from the cladding. This includes, but is not limited to, materials such as timber battens greater than or equal to 20mm thick, continuous thermal breaks such as polystyrene insulation sheeting, plastic strips or furring channels.
U-value	the rate of heat transfer through a window. The lower the U-value, the better the insulating ability.
Unconditioned	a zone within a dwelling that is assumed to not require heating and cooling based on standard occupancy assumptions.
Vertical shading features	provides shading to the building in the vertical plane and can be parallel or perpendicular to the subject wall/window. Includes privacy screens, other walls in the building (wing walls), fences, other buildings, vegetation (protected or listed heritage trees).
Window shading device	a device fixed to windows that provides shading e.g. window awnings or screens but excludes horizontal* or vertical shading features* (eg eaves and balconies)